AL/2023(2024)/20/E-I

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீட்சைத் திணைக்களம்
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2023 (2024)
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2023 (2024)
General Certificate of Education (Adv. Level) Examination, 2023 (2024)

තොරතුරු හා සන්නිවේදන තාක්ෂණය I
தகவல், தொடர்பாடல் தொழினுட்பவியல் I
Information & Communication Technology I

20 E I

පැය දෙකයි
இரண்டு மணித்தியாலம்
Two hours

Instructions:

- * *Answer all the questions.*
- * *Write your Index Number in the space provided in the answer sheet.*
- * *Instructions are also given on the back of the answer sheet. Follow those carefully.*
- * *In each of the questions 1 to 50, pick one of the alternatives from (1), (2), (3), (4), (5) which is correct or most appropriate and mark your response on the answer sheet with a cross (×) in accordance with the instructions given on the back of the answer sheet.*
- * *Use of calculators is not allowed.*

1. Which of the following statements are correct?

A – Word processors and spreadsheet software belong to the category of *utility software*.
B – A *compiler* is an example for a program translator.
C – It is illegal to use a proprietary software without obtaining its license.

(1) *A* only (2) *B* only (3) *C* only
(4) *A* and *B* only (5) *B* and *C* only

2. Personal information of students and their exam marks are input to a Student Information System. Marks for a subject range from 0 to 100. A student has to study a collection of compulsory and optional subjects and sit for the relevant examinations.
Which of the following are suitable data validations for the above system?

A – A *presence check* for the marks of all subjects taken/not taken by the student
B – A *range check* to check whether an input exam mark is within the range 0 and 100
C – A *data type check* to ensure that the input made for the telephone number of the student contains only digits

(1) *A* only (2) *B* only (3) *A* and *B* only
(4) *A* and *C* only (5) *B* and *C* only

3. The existing book management system in a school library is used with a computer, a monitor, a keyboard and a mouse. The school management wants to minimize the time taken presently for book lending/returning. Which of the following is most suitable for this purpose?

(1) Using a digitizer (2) Using an external hard disk
(3) Using a touch screen (4) Using a magnetic stripe reader
(5) Using bar code technology

4. Listed below are some phrases about the internal operation of three printers:

A – a moving print head striking an ink ribbon against the paper
B – toner attracting to what is printed on a cylinder which is then transferred to paper
C – nozzles spraying ink onto paper

Which of the following correctly matches *dot matrix*, *inkjet* and *laser* printers to the above phrases?

(1) *A* – dot matrix, *B* – laser, *C* – inkjet
(2) *A* – dot matrix, *B* – inkjet, *C* – laser
(3) *A* – inkjet, *B* – dot matrix, *C* – laser
(4) *A* – laser, *B* – dot matrix, *C* – inkjet
(5) *A* – laser, *B* – inkjet, *C* – dot matrix

[See page two

5. Which of the following will cause the CPU to execute a different set of instructions?
   A – a context switch
   B – an interrupt
   C – user selecting the shutdown option in the computer

(1) *A* only (2) *B* only (3) *C* only
(4) *A* and *B* only (5) All *A*, *B* and *C*

6. A program runs fastest when the data it requires are in the
(1) hard disk. (2) L1 cache. (3) L2 cache.
(4) magnetic tape. (5) main memory.

7. What is the correct binary equivalent of decimal 13.125_{10}?
(1) 1100.001 (2) 1100.100 (3) 1101.001 (4) 1101.100 (5) 1101.101

8. Which of the following are equivalent to octal 674_8?
   A – $110\ 111\ 100_2$
   B – 444_{10}
   C – $2BC_{16}$

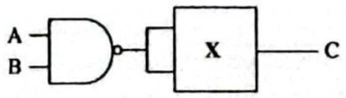
(1) *A* only (2) *A* and *B* only (3) *A* and *C* only
(4) *B* and *C* only (5) All *A*, *B* and *C*

9. The address of an instruction was shown as 5A1 in hexadecimal. What is that address in decimal?
(1) 41 (2) 1441 (3) 1457 (4) 2641 (5) 23056

10. A document contains 2048 characters including spaces and line-breaks. How many bits are needed to encode this document in ASCII also using the parity bits?
(1) 2048 (2) 2048 × 2 (3) 2048 × 7 (4) 2048 × 8 (5) 2048 / 8

11. What is the correct 2's complement binary representation of decimal -49_{10} using 8-bits?
(1) 00110001 (2) 01100010 (3) 10011110 (4) 11001111 (5) 11100010

12. Consider the following logic circuit in which X indicates a two-input logic gate.



Which of the following should X be so that when A = 0 and B = 1, the output C would be 0?
   I – a NAND gate
   II – a NOR gate
   III – an XOR gate

(1) I only (2) I and II only (3) I and III only
(4) II and III only (5) All I, II and III

13. Which of the following is the simplified form of the Boolean expression $X(\overline{X}+Y)$?
(1) X (2) Y (3) XY (4) $\overline{X}Y$ (5) X+Y

14. A program in execution in a computer is called a *process*. Which of the following is a possible state transition sequence of such a process?
(1) New → Ready → Running → Terminated
(2) New → Blocked → Terminated
(3) New → Ready → Blocked → Running → Terminated
(4) New → Running → Ready → Running → Terminated
(5) New → Blocked → Ready → Running → Terminated

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15. Amara powers on the computer and starts a spreadsheet application. Then he also opens a web browser. Which of the following are possible execution sequences on the processor of his computer?

(1) BIOS → OS → spreadsheet process → OS → web browser process → OS → ...
(2) BIOS → spreadsheet process → OS → web browser process → OS → spreadsheet process → ...
(3) BIOS → spreadsheet process → web browser process → OS → ...
(4) BIOS → OS → spreadsheet process → web browser process → OS → ...
(5) BIOS → OS → spreadsheet process → web browser process → spreadsheet process → web browser process → ...

16. Which of the following statements are true?

A – A *firewall* acts as a packet filter inspecting all the packets entering a network.
B – A malware that misleads the users by disguising itelf as a standard program is termed a Trojan Horse.
C – A strong password should have a combination of uppercase and lowercase letters, numbers and symbols of sufficient length.

(1) *A* only (2) *B* only (3) *C* only
(4) *A* and *B* only (5) All *A*, *B* and *C*

17. Which of the following statements are true?

A – One of the uses of encryption is to ensure confidentiality of transmitted data.
B – Every user needs to have a pair of dissimilar keys when using Asymmetric Key Encryption.
C – Users must share a common key when exchanging information using Symmetric Key Encryption.

(1) *A* only (2) *A* and *B* only (3) *A* and *C* only
(4) *B* and *C* only (5) All *A*, *B* and *C*

18. Which of the following is considered as an erroneously received byte in an *even parity system*?

(1) 01010101 (2) 10010011 (3) 10110010
(4) 11011001 (5) 11010111

19. Match the **Devices** labelled from **A** to **E** to the corresponding **Descriptions** labelled from **1** to **5**.

Device
A. Client
B. Hub
C. Router
D. Server
E. Switch

Description
1 – stores network programs and data files for the users to access
2 – a connecting device between Local Area Networks (LAN) and Wide Area Networks (WAN)
3 – when a message is received, this transmits it only on the port to which the destination computer is attached
4 – requests services and content from other computers
5 – when a message is received, this broadcasts it on all ports to all attached hosts

(1) **A – 1, B – 5, C – 4, D – 2, E – 3**
(2) **A – 2, B – 4, C – 3, D – 5, E – 1**
(3) **A – 3, B – 2, C – 1, D – 4, E – 5**
(4) **A – 4, B – 5, C – 2, D – 1, E – 3**
(5) **A – 5, B – 1, C – 2, D – 3, E – 4**

[*See page four*

20. Select the answer containing the correct replacements for Ⓐ and Ⓑ in the following paragraph:
In the Internet, a host is identified by its IP address. In IPv4, each IP address consists of ___Ⓐ___ bits to identify a host. The newer version named IPv6 consists of ___Ⓑ___ bits in an IP address.

(1) Ⓐ = 32, Ⓑ = 48 (2) Ⓐ = 32, Ⓑ = 128 (3) Ⓐ = 48, Ⓑ = 32
(4) Ⓐ = 48, Ⓑ = 128 (5) Ⓐ = 128, Ⓑ = 32

21. Which of the following statements regarding DNS (Domain Name System) are correct?
A – It maps web addresses to IP addresses and vice versa.
B – HTTP uses the services provided by the DNS.
C – DNS maintains a hierarchy of domain names.
(1) *A* only (2) *A* and *B* only (3) *A* and *C* only
(4) *B* and *C* only (5) All *A*, *B* and *C*

22. Which of the following statements regarding IP addresses are correct?
A – In class C networks, first octet value ranges from 192 through 223.
B – IPv4 can assign addresses up to 4 million devices.
C – 192.168.0.0 – 192.168.255.255 is a private IP address range.
(1) *A* only (2) *B* only (3) *C* only
(4) *A* and *B* only (5) *A* and *C* only

23. If Suresh wants to send an encrypted message to be read only by Amara using *asymmetric key encryption*, then
(1) Suresh should encrypt his message using his public key.
(2) Suresh should encrypt his message using his private key.
(3) Suresh should encrypt his message using Amara's public key.
(4) Suresh should encrypt his message using Amara's private key.
(5) Suresh should encrypt his message using both private and public keys of Amara.

24. Choose the option containing most suitable deployment types for the following systems:
A – A new system to replace an existing air traffic control system at an airport
B – A system for the customers of an island-wide supermarket chain to order goods online
C – A system for the public to enter comments regarding the service experienced by them at an office
(1) *A* – direct, *B* – direct, *C* – parallel
(2) *A* – direct, *B* – pilot, *C* – parallel
(3) *A* – parallel, *B* – pilot, *C* – direct
(4) *A* – parallel, *B* – parallel, *C* – parallel
(5) *A* – parallel, *B* – parallel, *C* – pilot

25. Which of the following is a *non-functional requirement* for an e-commerce site?
(1) Being able to add items to the shopping cart
(2) Being able to make payments online
(3) Being able to view the items based on item category
(4) Each item to be shown with a small image and a description
(5) The e-commerce site to be accessible through popular web browsers

26. During which of the following is an application tested by its developers in a setting that closely resembles its intended deployment hardware, software, and network configuration environment?
(1) Acceptance testing (2) Integration testing (3) Parallel testing
(4) System testing (5) Unit testing

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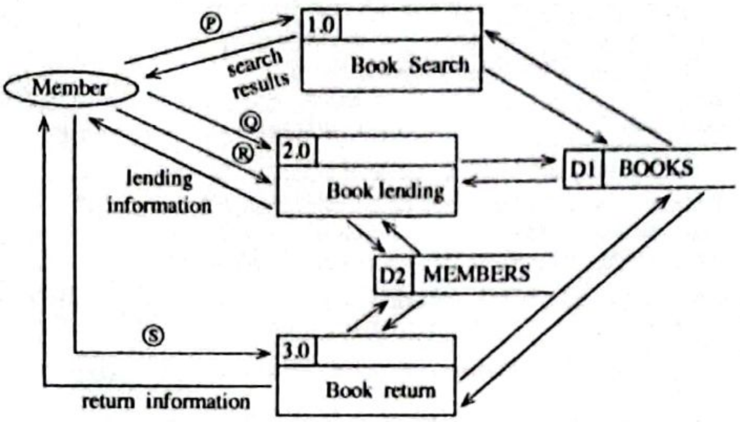
27. A company considers developing a new software application for its use. The application is expected to re-engineer internal processes, improve collaboration, and enhance productivity. However, during the feasibility analysis, it was identified that the new software may face some resistance from employees who are accustomed to the existing processes. Which component of the feasibility study would have helped to get that information?

(1) economic feasibility (2) legal feasibility
(3) operational feasibility (4) schedule feasibility
(5) technical feasibility

28. Select the option which includes most suitable replacements for the labels Ⓟ to Ⓢ in the following Data Flow Diagram of a library management system.



(1) Ⓟ – keyword, Ⓠ – member ID, Ⓡ – book details, Ⓢ – book details
(2) Ⓟ – keyword, Ⓠ – keyword, Ⓡ – book details, Ⓢ – member ID
(3) Ⓟ – keyword, Ⓠ – keyword, Ⓡ – book details, Ⓢ – keyword
(4) Ⓟ – member ID, Ⓠ – keyword, Ⓡ – member ID, Ⓢ – member ID
(5) Ⓟ – member ID, Ⓠ – member ID, Ⓡ – book details, Ⓢ – book details

29. Which of the following is **incorrect** about the waterfall model of software development?

(1) It allows developers to collect and implement requirements throughout the project.
(2) It is not an iterative model.
(3) It is suitable for software with well defined requirements.
(4) It is easy to estimate the resources needed for a project.
(5) No working product is available until the latter stages of the project.

30. In addition to the required features, which of the following should also be considered when a government institution selects a Commercial Off-The-Shelf (COTS) software to be implemented island wide?

A – cost to deploy, maintain, upgrade and modify
B – ease of integration with existing systems
C – after sales service from the vendor

(1) *A* only (2) *A* and *B* only (3) *A* and *C* only
(4) *B* and *C* only (5) All *A*, *B* and *C*

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31. Match the given **entity attributes** labelled from **A** to **D** to the corresponding **descriptions** labelled from **1** to **4**.

	Entity attribute
A	Composite attribute
B	Simple attribute
C	Multivalued attribute
D	Derived attribute

	Description
1	an attribute that cannot be broken down into smaller components
2	an attribute that can be broken down into component parts
3	an attribute whose values can be calculated from related attribute values
4	an attribute that may take more than one value

(1) A-2, B-1, C-3, D-4
(2) A-2, B-1, C-4, D-3
(3) A-3, B-4, C-2, D-1
(4) A-4, B-2, C-3, D-1
(5) A-4, B-3, C-1, D-2

32. Consider the following **Employee** Relation:

Employee_ID	Employee_Name	Salary
1001	John	60000
1002	Hari	55000
1003	Mahas	70000
1004	Sarath	65000
1005	Rajah	75000

What would be the output of the following SQL query when it is applied on the **Employee** relation?

```
SELECT COUNT(*)
FROM Employee
WHERE Salary > ANY (SELECT Salary FROM Employee);
```

(1) 3 (2) 4 (3) 5 (4) 6 (5) 10

33. Consider the given SQL statements to create two database tables named **LENDING** and **STUDENT**:

```
CREATE TABLE LENDING
 (BOOK_NUMBER VARCHAR(10) NOTNULL,
  BOOK_NAME VARCHAR(20) NOTNULL,
  AUTHOR VARCHAR(25) NOTNULL,
  DESCRIPTION VARCHAR(75) NOTNULL,
  ISSUED_DATE DATE,
  STUDENT_ID CHAR(5) NOTNULL,
  PRIMARY KEY(BOOK_NUMBER));

CREATE TABLE STUDENT
 (STUDENT_ID CHAR(5) NOTNULL,
  NAME VARCHAR(25) NOTNULL,
  BIRTHDAY DATE NOTNULL,
  ADDRESS VARCHAR(25) NOTNULL,
  PROVINCE CHAR(10),
  PRIMARY KEY(STUDENT_ID));
```

Which of the following statements are correct?

A – STUDENT_ID is a foreign key in the LENDING table.
B – It is compulsory to input data to the DATE data type fields in both tables.
C – STUDENT_ID can contain only five English letters.

(1) *A* only
(2) *A* and *B* only
(3) *A* and *C* only
(4) *B* and *C* only
(5) All *A*, *B* and *C*

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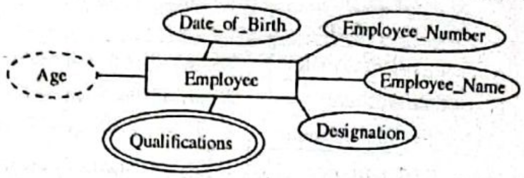
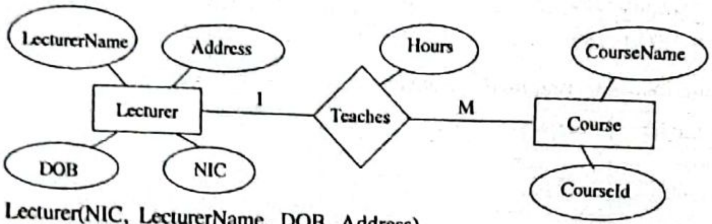
34. When the **Employee** entity of the following diagram is represented in a database which of the following should **not** be included?

(1) Date_of_Birth
(2) Designation
(3) Employee_Name
(4) Employee_Number
(5) Qualifications



35. Which of the listed relations will be obtained if the following ER diagram is correctly mapped into the relational model?



A – Lecturer(NIC, LecturerName, DOB, Address)
B – Lecturer(NIC, LecturerName, DOB, Address, CourseId)
C – Teaches(NIC, CourseId, Hours)
D – Course(CourseId, CourseName, Hours, NIC)

(1) A and B only
(2) A and C only
(3) A and D only
(4) B and C only
(5) A, C and D only

36. Which of the following gives a correct matching between ER diagram components and the relational model?

(1) Entity → Field, Attribute → Table, Unique attribute → Primary key, Multivalued attribute → Table
(2) Entity → Table, Attribute → Field, Unique attribute → Primary key, Multivalued attribute → Table
(3) Entity → Table, Attribute → Field, Unique attribute → Table, Multivalued attribute → Primary key
(4) Entity → Table, Attribute → Primary key, Unique attribute → Primary key, Multivalued attribute → Table
(5) Entity → Table, Attribute → Table, Unique attribute → Primary key, Multivalued attribute → Primary key

- Consider the following relations to answer questions **37** and **38**.

adviser (<u>adId</u>, adName, adGender, adNIC, adPhone)
farmer (<u>farmerId</u>, farmerName, farmerAddress, farmerPhone)

task (<u>taskId</u>, taskName, farmerId, startDate, endDate)
advises (<u>adId</u>, <u>taskId</u>, startDate, endDate)

Note: adNIC – The National Identity Card number of an adviser

37. Which of the following statements are correct?

A – One farmer can have many tasks.
B – One adviser can advise many tasks.
C – For one task, a farmer can have many advisers.

(1) *A* only
(2) *A* and *B* only
(3) *A* and *C* only
(4) *B* and *C* only
(5) All *A*, *B* and *C*

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38. Which of the following statements are correct with respect to the given relations?
    A – All relations are in 3rd normal form.
    B – The startDate attribute in the **task** relation is a derived attribute.
    C – adNIC is a candidate key in the **adviser** relation.

(1) *A* only (2) *A* and *B* only (3) *A* and *C* only
(4) *B* and *C* only (5) All *A*, *B* and *C* only

39. What would be the output of the following Python code, if a = 10, b = 4, and c = 7?

```
ans=a%b+c//(a-b)
print(ans)
```

(1) 3 (2) 5 (3) 7 (4) 9 (5) 11

40. What would be the value of the 'result' variable after executing the following Python code?

```
def  func1(a,b):
       return  a+b

def  func2(a,b):
       return  a*b

result  =  func1(3,func2(2,4))
```

(1) 11 (2) 12 (3) 14 (4) 15 (5) 20

41. What would be the output of the following Python code, when it gets executed?

```
def  modify_string(input_string):
       input_string+=" World"
text="Hello"
modify_string(text)
print(text)
```

(1) Hello (2) Hello Hello
(3) Hello World (4) World
(5) World Hello

42. What would be the output of the following Python code?

```
original_list=[1,2,3,4,5]
new_list=original_list.copy()
new_list.clear()
original_list.append(6)
print(original_list)
print(new_list)
```

(1) [ ]
[ ]

(2) [6]
[ ]

(3) [6]
[6]

(4) [1, 2, 3, 4, 5, 6]
[ ]

(5) [ ]
[1, 2, 3, 4, 5, 6]

[See page nine

43. How many '*'s does this program output?

```
i=7
while i>0:
  i-=3
  print('*')
  if i<=2:
      break
else:
   print('*')
```

(1) 1 (2) 3 (3) 5 (4) 7 (5) 9

44. Which of the data structures among *Dictionary, List* and *Tuple* in Python could be used to store a collection of key-value pairs where the keys must be unique?

(1) Dictionary only (2) List only (3) Tuple only
(4) Dictionary and List only (5) List and Tuple only

45. What would be the output of the following python code?

```
for i in range(1, 4):
    for j in range(1, i + 1):
        print(j * i, end=' ')
    print()
```

(1)
```
1
2 2
3 3 3
```
(2)
```
1
2 4
3 6
```
(3)
```
1
2 4
3 6 9
```
(4)
```
1 2 3
2 4 6
3 6 9
```
(5)
```
1 2
2 4 6
3 6 9 12
```

46. Consider the following code fragment in an HTML file:

```
<style>
    body {
        color: yellow;
        font-family: Arial, Cambria;
    }
    .highlight {
        color: red;
    }
</style>
```

What happens if one applies the class 'highlight' to a <div> element within <html> and </html> tags in the above file?

(1) The <div> element's text will turn red.
(2) The <div> element's text will turn yellow.
(3) The <div> element's font size will increase.
(4) The <div> element's font type will change to Cambria.
(5) The <div> element's border colour will change to red.

47. Which of the following statements regarding Search Engine Optimization (SEO) are correct?

A – Meta tags on web pages help SEO.
B – It increases the visibility of a web page in search engines.
C – Powerful computers should be used to create SEO friendly web pages.

(1) *A* only (2) *A* and *B* only (3) *A* and *C* only
(4) *B* and *C* only (5) All *A*, *B* and *C* only

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48. Consider the following HTML code line related to a form:

```
<form method="post" action="process.php">
```

The "action" attribute in it

(1) specifies the data type of the form.
(2) specifies the server file that handles the data in the form.
(3) controls the form's alignment on the web page.
(4) declares the form as a PHP script.
(5) shows the process.php file on the screen.

49. Saman's father is a carpenter. He wants to showcase his father's work on a website. Which of the following hosting options should Saman use in order to do it with a price that he can afford?

(1) Hosting it on a server that presents other websites also (shared hosting)
(2) Hosting it on a Virtual Private Server (VPS)
(3) Hosting it on a server dedicated to Saman (dedicated hosting)
(4) Using an e-Commerce website
(5) Using the services of a well known cloud service provider

50. What is the primary role of a sensor in an IoT device?

(1) To provide outputs and change a state of the environment
(2) To ensure interoperability of devices
(3) To detect a state change in the environment
(4) To make decisions based on predetermined rules
(5) To generate graphics for the user interface

* * *

3 Paper I answers

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீட்சைத் திணைக்களம்

අ.පො.ස. (උ.පෙළ) විභාගය / க.பொ.த. (உயர் தர)ப் பரீட்சை - 2023 (2024)

විෂය අංකය / பாட இலக்கம் **20**

විෂය / பாடம் Information and Communication Technology

ලකුණු දීමේ පටිපාටිය / புள்ளி வழங்கும் திட்டம்

I පත්‍රය / பத்திரம் I

ප්‍රශ්න අංකය / வினா இல.	පිළිතුරු අංකය / விடை இல.	ප්‍රශ්න අංකය / வினா இல.	පිළිතුරු අංකය / விடை இல.	ප්‍රශ්න අංකය / வினா இல.	පිළිතුරු අංකය / விடை இல.	ප්‍රශ්න අංකය / வினா இல.	පිළිතුරු අංකය / விடை இல.	ප්‍රශ්න අංකය / வினா இல.	පිළිතුරු අංකය / விடை இல.
01.	5	11.	4	21.	5	31.	2	41.	1
02.	5	12.	5	22.	5	32.	2	42.	4
03.	5	13.	3	23.	3	33.	1	43.	2
04.	1	14.	1	24.	3	34.	5	44.	1
05.	5	15.	1	25.	5	35.	3	45.	3
06.	2	16.	5	26.	4	36.	2	46.	1
07.	3	17.	5	27.	3	37.	2, 5	47.	2
08.	2	18.	4	28.	1	38.	3	48.	2
09.	2	19.	4	29.	1	39.	1	49.	1
10.	4	20.	2	30.	5	40.	1	50.	3

O විශේෂ උපදෙස් / விசேட அறிவுறுத்தல் :

එක් පිළිතුරකට / ஒரு சரியான விடைக்கு 01 ලකුණු / புள்ளி வீதம்

මුළු ලකුණු / மொத்தப் புள்ளிகள் 1 × 50 = 50

AL/2023(2024)/20/E-II 09799

Department of Examinations, Sri Lanka

General Certificate of Education (Adv. Level) Examination, 2023(2024)

Information & Communication Technology II **20 E II**

Three hours

Additional Reading Time - 10 minutes

Use additional reading time to go through the question paper, select the questions you will answer and decide which of them you will prioritise.

Index No. : ....

Important:

- *This question paper consists of* **13** *pages.*
- *This question paper comprises of two parts,* **Part A** *and* **Part B.** *The time allotted for both parts is* **three hours.**
- *Use of calculators is not allowed.*

PART A — Structured Essay: (pages 2 - 7)

- *Answer all the questions on this paper itself. Write your answers in the space provided for each question. Note that the space provided is sufficient for your answers and that extensive answers are not expected.*

PART B — Essay: (pages 8 - 13)

- *This part contains* **six** *questions, of which, four are to be answered. Use the papers supplied for this purpose.*
- *At the end of the time allotted for this paper, tie the two parts together so that* **Part A** *is on top of* **Part B** *before handing them over to the Supervisor.*
- *You are permitted to remove only* **Part B** *of the question paper from the Examination Hall.*

For Examiners' Use Only

For the Second Paper

Part	Question No.	Marks
A	1	
	2	
	3	
	4	
B	5	
	6	
	7	
	8	
	9	
	10	
Total		

Final Marks

In numbers	
In words	

Code Number

Marking Examiner 1	
Marking Examiner 2	
Marks checked by:	
Supervised by:	

[see page two

Part A – Structured Essay

*Answer **all four** questions on this paper itself.*

Do not write in this column

1. (*a*) Draw the expected output of the following HTML code segment when rendered by a web browser.

```
<html>
<body>
<ul  style="list-style-type:none;">
       <li>Cricket</li>
       <li>Football</li>
       <li>Hockey</li>
</ul>
<html>
<body>
```

Note: Consider the following dotted line box as the display area of the browser.

(*b*) A registration form for a speech competition and its labeled HTML source are given in Figures 1.1 and 1.2 respectively.

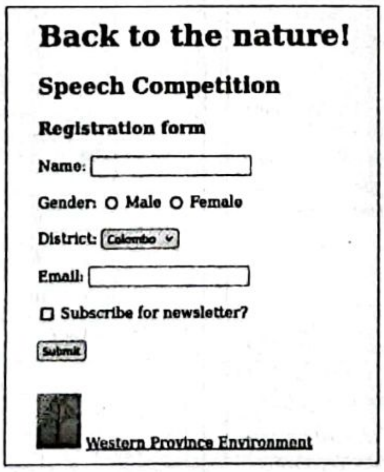


Figure 1.1

[see page three

Do not write in this column

```
<html>
<Ⓐ>Back to the nature!</Ⓐ>
<Ⓑ>Speech Competition</Ⓑ>
<h3>Registration form</h3>

<form method="Ⓒ" Ⓓ="./action_page.php">
      <label for="name">Name:</label>
      <input type="Ⓔ" name="name"><br><br>

      <label for="gender">Gender:</label>
      <input type="Ⓕ" name="gender" id="male" value="male">
      <label for="male">Male</label>
      <input type="Ⓕ" name="gender" id="female" value="female">
      <label for="female">Female</label> <br><br>

      <label for="Ⓖ">District: </label>
      <Ⓗ name="district" id="district">
             <option value="colombo">Colombo</option>
             <option value="gampaha">Gampaha</option>
             <option value="kalutara">Kalutara</option>
      </Ⓗ><br><br>
      <label for="email">Email:</label>
      <input type="email" name="email"><br><br>

      <input type="Ⓘ" name="newsletter" id="newsletter">
      <label for="newsletter">Subscribe for newsletter?</label><br><br>

      <input type="Ⓙ" value="Submit">
</form>
<br>
  <Ⓚ="wpeLogo.jpg" alt="Ⓛ" width="50" height="60">
  <Ⓜ="https://www.wpe.lk" title="Ⓝ">Western Province Environment</a>
</html>
```

Figure 1.2

For each of the labels Ⓐ to Ⓝ in the HTML code in Figure 1.2, choose a suitable replacement from the given list. In the answer table, write down the **number** of the replacement for each label.

List:

1: action	2: a href	3: caption	4: checkbox	5: district
6: font	7: h1	8: h2	9: h3	10: head
11: img src	12: More details	13: name	14: post	15: radio
16: select	17: submit	18: text	19: th	20: WPE logo

Answer table:

Ⓐ :	Ⓑ :	Ⓒ :	Ⓓ :	Ⓔ :	Ⓕ :	Ⓖ :
Ⓗ :	Ⓘ :	Ⓙ :	Ⓚ :	Ⓛ :	Ⓜ :	Ⓝ :

[see page four

(*c*) The action_page.php file mentioned in the given code of Figure 1.2 is shown below.

Do not write in this column

```
<?php
$servername = "localhost";   $username = "root";   $password = "";
$dbname = "environment";
// Create a connection
$conn = new mysqli($servername, $username, $password, $dbname);

// Section P
$name = $_POST['name']; $gender = $_POST['gender']; $district = $_
POST['district']; $email = $_POST['email']; $newsletter = $_POST['news-
letter'];
// section P end
// Section Q
$sql = "INSERT INTO applicants (name, gender, district, email,
newsletter) VALUES ('$name', '$gender', '$district', '$email',
'$newsletter')";
// section Q end

if ($conn->query($sql) === TRUE) {
      echo "Data inserted successfully!";
} else {
      echo "Error: " . $sql . "<br>" . $conn->error;
}

// Close the connection
mysqli_close($conn);
?>
```

Write down the purpose of section **P** and the purpose of section **Q**.

P : ....

Q : ....

2. (*a*) A simple and high-level view of the data lifecycle consists of three steps. Write the 2nd and the 3rd steps of the data life cycle.

- 1st step is the creation of data
- 2nd step is ....
- 3rd step is ....

(*b*) (i) Modern Artificial Intelligence relies on large amounts of data, which are often managed with cloud-based storage solutions. What is the cloud computing service model used here?

....

(ii) Quantum computers, although seen as a promising type of computing machines for the future, are still expensive to own, operate and maintain. Suggest a technical approach to make the computing power of the quantum computers accessible to the public users as per their needs, at an affordable price.

....

....

(*c*) For the box in each of the following statements, select a suitable replacement from the given list and write the number of the selected replacement in the box.

List : {**1** - B2B, **2** - C2B, **3** - G2C, **4** - a payment gateway, **5** - a reverse auction, **6** - a virtual storefront, **7** - a web portal, **8** - an online auction, **9** - an online marketplace}

[see page five

Do not write in this column

(i) A web-based platform that provides a single point of access to a range of information from different sources, is known as ☐.

(ii) When one requests to renew his/her vehicle revenue license and pays online for it through the official website, he/she is performing an e-commerce transaction of ☐ type.

(iii) The ABC e-commerce company does not allow buyers to explore competitor products from other sellers within its website. ABC website operates as ☐.

(iv) In ☐ buyers bid for the prices at which they are willing to buy a given product or service.

(v) An online shopping website is suitable to be connected to ☐.

(*d*) (i) Your friend thinks the digital divide is a tool used for arithmetic division. Briefly explain to your friend what the digital divide is.

....

....

....

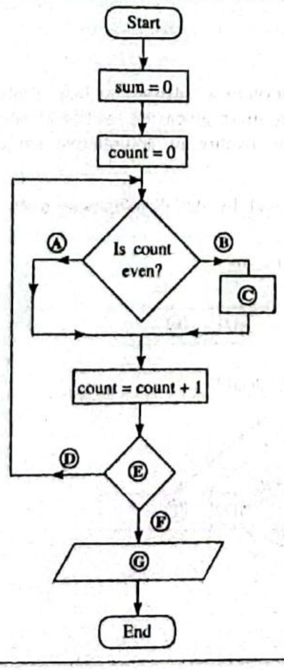
(ii) E-waste is becoming a major environmental problem in Sri Lanka. Suggest a step that we can take to reduce the environmental impact of our e-waste.

....

....

....

3. (*a*) Write down the most suitable replacements for labels Ⓐ to Ⓖ in the following flowchart which is drawn to calculate and display the sum of first ten even numbers.



Ⓐ : ....

Ⓑ : ....

Ⓒ : ....

Ⓓ : ....

Ⓔ : ....

Ⓕ : ....

Ⓖ : ....

[see page six

Do not write in this column

(*b*) (i) What is the output of the following Python code?

```
def func(n):
    MyNumber=[]
    for i in range(4,n+1):
        if i%2==0:
            MyNumber.append(i)
    print(MyNumber)
func(30)
```

....

(ii) Write down the output in the above Python code when the condition `if i%2==0:` is changed to `if i%2 != 0:`

....

(*c*) Write down the replacements for the labels of the following Python code which has been written to find the largest of a set of integers.

```
def findlargest(myList):
    largest = Ⓐ
    for i in Ⓑ:
        if i>Ⓒ
            largest = Ⓓ
    print("largest value is",Ⓔ)
list1=[4,6,24,12,8,94,22]
findlargest(Ⓕ)
```

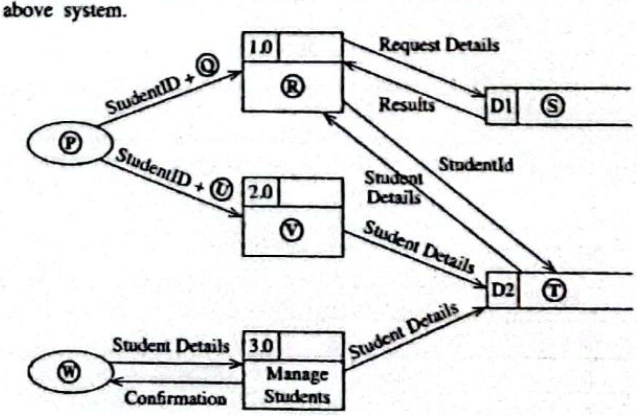
Ⓐ – .... Ⓑ – ....

Ⓒ – .... Ⓓ – ....

Ⓔ – .... Ⓕ – ....

4. A team in the school IT society has been requested to develop a software to help students reserve computers in the school laboratory. The students are to be given the facility to update their information. The administrator should be given the facility to add/remove students to/from the system.

(*a*) The following is the data flow diagram (DFD) prepared by the development team for the above system.



[see page seven

A numbered list of replacements for the labels Ⓟ to Ⓦ are given below. Write down in the relevant box the **number** of the most suitable replacement for each of the labels given in the DFD.

List: {**1** – Administrator, **2** – Handle request, **3** – Reservations, **4** – Request details, **5** – Student, **6** – Students, **7** – Student details, **8** – Update student details}

Ⓟ – ☐ Ⓠ – ☐ Ⓡ – ☐ Ⓢ – ☐

Ⓣ – ☐ Ⓤ – ☐ Ⓥ – ☐ Ⓦ – ☐

(*b*) The computers are to be made available for students only for 30-minute slots between 8 am – 5 pm on weekends. However a student is allowed to reserve only a maximum of two 30-minute slots per weekend.

Write down **one** functional requirement with respect to computer reservations.

....

....

(*c*) Give **one** technical aspect that the development team should check when conducting the technical feasibility study of this project.

....

....

(*d*) The *waterfall model* is suggested for the above development. Why is a proper requirement analysis critical in this project to ensure its timely completion?

....

....

(*e*) Three students are to develop the *reservation*, *update student details* and *manage students* modules separately. The IT teacher had taught different types of software testing. What is meant by "integration testing" in this system?

....

....

(*f*) The IT teacher suggests the team doing a *direct deployment* of this software. Give **one** reason as to why the teacher did not suggest a *parallel deployment*.

....

....

(*g*) One member of the IT society wants a COTS (Commercial-Off-The-Shelf) software to be considered for this system instead of developing it. Give **one** reason as to why the team should reject it.

....

....

* *

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[see page eight

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இலங்கைப் பரீட்சைத் திணைக்களம்
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2023(2024)
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2023(2024)
General Certificate of Education (Adv. Level) Examination, 2023(2024)

තොරතුරු හා සන්නිවේදන තාක්ෂණය II
தகவல், தொடர்பாடல் தொழினுட்பவியல் II
Information & Communication Technology II

20 E II

Part B

* *Answer any four questions only.*

5. (*a*) A circuit with three inputs (A, B, C) and one output (Z) is to be designed. The output should be equal to 1 when the binary value combination of the three inputs is either 1, 3 or 6. The output should be 0 for other cases.

(i) Draw the complete truth table for the above circuit.

(ii) Complete the Karnaugh map relevant to the above circuit according to the following format:

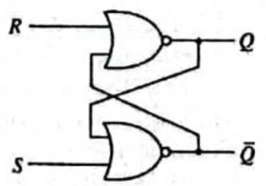
C \ AB	00	01	11	10
0				
1				

(iii) Using the Karnaugh map, derive the most simplified product-of-sums (POS) expression for the output Z. Show the loops clearly on the Karnaugh map.

(iv) Draw a logic circuit for the simplified expression derived in (iii) by only using NOR gates assuming that the complemented inputs $\bar{A}$, $\bar{B}$ and $\bar{C}$ are also available.

(*b*) Using Boolean Algebra show that $\bar{A}C+\bar{A}B+A\bar{B}C+BC$ is equivalent to $C+\bar{A}B$.

(*c*) Consider the flip flop circuit shown on the right.



(i) Assume that the S input is 1 and the R input is 0. What will be the output at Q?

(ii) What will be the output at Q if the S input is now made 0?

(iii) What will be the output at Q when the R input is now made 1?

6. (*a*) Draw a sketch to show how a file server (FS), a printer (P), a switch (S) and two computers (C1 and C2) should be connected in a *star topology*.

(*b*) A port number is also used along with an IP address in a network communication. Why?

(*c*) Consider a subnet with the network address 192.168.56.128/26.

(i) Write an example IP address that can be assigned to a host attached to this subnet (in dotted decimal notation).

(ii) Write the first and the last usable host addresses in this network (In dotted decimal notation).

(iii) How many host addresses are available for use in this subnet?

[see page nine

(*d*) Suppose an Internet Service Provider owns the 192.168.56.32/26 IP address block. Assume that the provider wants to create four subnets namely, Subnet A, Subnet B, Subnet C and Subnet D from this address block with each subnet having the same number of IP addresses.

(i) Write the subnet mask of the above given IP address block in dotted decimal notation.

(ii) Write the number of host bits needed to create the required number of subnets.

(iii) Once subnetting is done, fill in the following table.

Subnet	Network address	First usable IP address	Last usable IP address	Broadcast address
Subnet A				
Subnet B				
Subnet C				
Subnet D				

(*e*) (i) Write two functions of a proxy server in a computer network.

(ii) Write two properties of MAC addresses assigned to devices connected to a network.

7. (*a*) Assume that you are given an Arduino UNO board (Figure 7.1) along with the following items:

- Passive Infrared Sensor (PIR) for motion detection (Figure 7.2)
- Sensor for ambient light detection
- LEDs, Resistors, and a Power supply

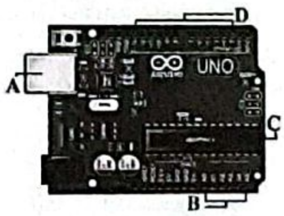


Figure 7.1

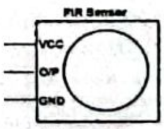


Figure 7.2

(i) Identify the parts marked as A, B, C, and D in Figure 7.1 and briefly explain each of their functionalities.

(ii) Assume that you want to build an IoT setup that switches an LED light on when motion is detected. It is further required to switch on this LED only during night time. Draw a schematic diagram connecting the Arduino board and the items given above as necessary in order to build this setup.

(*b*) An e-commerce warehouse automation system includes a set of agent-based robots which move ordered goods to their respective dispatch areas to start relevant shipments.

The Figure 7.3 shows the latter part of this system. A Quality Control (QC) Officer inspects the goods of each order as it passes on a conveyor belt and confirms to a software system (Delivery Handler Agent) that the order has passed QC. The Delivery Handler Agent directs the package to a mobile robot at the loading area. The robot agent reads the package barcode to determine the appropriate dispatch area. It then navigates the robot to the relevant dispatch area, scanning the path and avoiding obstacles while on the move. The Dispatch Handler Agent, another software, validates each package at the dispatch areas and informs the Dispatch Officer to confirm its decision. The Dispatch Officer can override Dispatch Handler decisions if needed and directs the confirmed packages to the postal division.

[see page ten

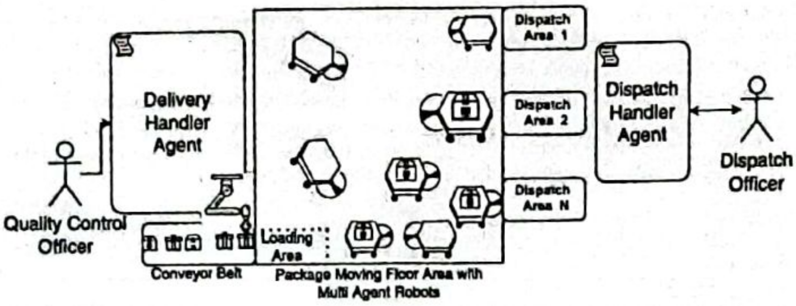


Figure 7.3

(i) Software Agents demonstrate certain characteristics which make their behaviour unique. Briefly explain the following two characteristics of a software agent:
(a) autonomous
(b) cooperative

(ii) Name a self-autonomous agent and a user agent in the given example.

(iii) If the set of multi-agent robots behave satisfying only the autonomous characteristic but fails to cooperate, write down one of the most likely observations that will be seen during their operation.

(iv) If this system is redesigned by replacing the multi-agent behaviour with centralized control and a broker agent for communication, identify one main change that will be seen with respect to each of the following:
(a) Control of the robot mobility
(b) Decision making process (relevant to moving packages from loading area to dispatch areas)

(v) Draw a *box and arrow diagram* for the new solution with centralized control, mentioned in (iv), above.
(Note: A box and arrow diagram uses boxes to show system components and arrows to show connections between those components)

8. (*a*) Write the output of the Python code given in Figure 8.1.

```
def function1(str):
   newstr = ''''
   for character in str:
        if character in 'aeiouAEIOU':
             newstr += '*'
        else:
             newstr += character
   return newstr
str1 = "LibrAry"
str2 = function1(str1)
print(str2)
```

Figure 8.1

(*b*) The function in Figure 8.2 uses the bubblesort algorithm to sort a given list of numbers into ascending order. Write down the suitable replacements for the labels P-U to complete the code.

```
def bubbleSort(nList):
   for pNumber in range(P,Q,R):
        S:
          if nList[i]>nList[i+1]:
              temp = nList[i]
              T
              U
```

Figure 8.2

[see page eleven

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(c) An estate owner wants a program to determine the minimum currency note combination needed to make the pay of each employee. (E.g., Rs. 40,000 should be paid using eight notes of Rs. 5000 and not four hundred notes of Rs. 100). The program should also output the currency requirement for all employees. The program should use the employees.txt file which contains employee pay details. Each line in it contains an employee's name and net pay.
A Python program written for this purpose is shown in Figure 8.3. A sample employees.txt file and the program's output for that file are shown in Figure 8.4.

(i) Write down the suitable replacements for the ten labels A-J in the program given in Figure 8.3.

```
# currency notes used in Sri Lanka
notes = [5000,1000,500,100,50,20]
# total notes required from each currency note type
totals = [0,0,0,0,0,0]

file = A('employees.txt','r')

while True:
    required = [0,0,0,0,0,0] # notes required for employee

    line = file.readline()
    if B line:
        C

    empDetails = line.split()
    netpay = int(float(D))
    if netpay < 0:
        continue

    print("\n")
    print(empDetails[0]," Net pay =",netpay)
    topay = netpay
    i = 0
    while topay > 0:
        required[i] = E
        totals[i] = totals[i] + F
        topay = G
        H

    # print employee netpay breakdown
    for i in range(0, len(required)):
        print("Rs.",notes[i],":", I)
J

print("\nTOTAL REQUIREMENT:")
for i in range(0, len(totals)):
    print("Rs.",notes[i],":",totals[i])
```

Figure 8.3

Example 'employees.txt' file:

```
Raj 40120
Niranjala 51670
```

Program's output for that file:

```
Raj  Net pay = 40120
Rs. 5000 : 8
Rs. 1000 : 0
Rs. 500 : 0
Rs. 100 : 1
Rs. 50 : 0
Rs. 20 : 1

Niranjala  Net pay = 51670
Rs. 5000 : 10
Rs. 1000 : 1
Rs. 500 : 1
Rs. 100 : 1
Rs. 50 : 1
Rs. 20 : 1

TOTAL REQUIREMENT:
Rs. 5000 : 18
Rs. 1000 : 1
Rs. 500 : 1
Rs. 100 : 2
Rs. 50 : 1
Rs. 20 : 2
```

Figure 8.4

(ii) The net pay of employees in this estate, does not contain cents. However, what practical problem with respect to the net pay inputs exists in this code? What modifications will you do to fix that problem?

[see page twelve

9. (*a*) Consider the following requirements relevant to a database that is expected to manage divisions, officers and tasks in an office.

The office consists of a number of divisions. Each division has a unique name. The division may have several locations. A division handles a number of tasks each of which has a unique number, a name and a date in which the task was assigned to the division. Each officer's name (consisting of a first name and a surname), NIC (National Identity Card) number, address and phone number is to be stored. An officer is assigned to one division but may work on several tasks which may not be controlled by the same division. Each division is managed by one of its officers and the starting date in which the officer started managing the division is stored.

Draw an ER digram for this application showing the entities, attributes and relationships. Underline primary keys.

(*b*) Write **two** advantages of converting a database table into a normal form.

(*c*) Consider the following **Show** table related to theatres and the movies that they screen.

Theatre	Movie	Day	Time	Screen	Year
Sarasi	MI - 4	Wednesday	10:00	S_1	2022
Sarasi	MI - 4	Wednesday	15:00	S_1	2022
Palazzo	Spider man	Friday	10:00	S_2	2019
Palazzo	Avengers	Friday	10:00	S_1	2019
Vega	Iron man	Thursday	10:00	S_1	2020

Note:

- A theatre can screen more than one movie at the same time on different screens.
- *Year* field gives the year in which the relevant film was released.

(i) In which normal form does the Show table exist? Justify your answer.

(ii) Convert the **Show** table to its next normal form.

(*d*) Consider the following **Employee** table:

Emp_ID	Emp_Name	DoB	Department	Designation	DoJ	Salary
E110	Saman	15/10/1970	Bio Technology	Professor	12/04/2001	145000
E111	Kumar	25/05/1980	Mechanical	Assistant Professor	02/05/2006	100000
E115	Raja	10/08/1982	Engineering	Assistant Professor	05/05/2001	98000
E114	Jennifer	11/09/1975	Engineering	Assistant Professor	03/06/2001	197000
E117	Ismail	15/05/1979	Civil	Assistant Professor	10/05/2005	103000

(i) Write the most suitable SQL statement to create the **Employee** table with a suitable primary key.

(ii) Write the required SQL statement to insert the record for the following employee:
Emp_ID = E119, Emp_Name = "John", DoB = "15/06/1971", Department = "IT", Designation = "Professor", DoJ = "15/07/2001", Salary = 107000

(iii) Write the output obtained by applying the following SQL query:

```
SELECT Emp_ID, Emp_Name
FROM Employee
WHERE Salary>103000;
```

(iv) Write the appropriate SQL query to find the names of all employees who work in the "Civil" department.

[*see page thirteen*

10. (*a*) (i) What is the repeating cycle that a processor in a computer is involved in since the computer is started till it is shutdown?

(ii) Which program's instructions get executed in the processor of a computer during a *context switch*?

(iii) A *register* is a group of binary cells suitable for holding binary information and is constituted by a collection of flip-flops. How many flip-flops are needed to make an n-bit register?

(*b*) A user runs the following Python codes on a computer. The code on left prints the lines of a file on the screen while the other code does an average computation.

fileReader.py	average.py

A = input ("Enter filename")
f1 = open(A, "r")
for line in f1:
    print(line)
f1.close()
	total = 0
	for num in range (10000):
	    total += num
	average = total / 10000
	print(average)

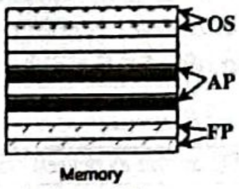


Figure 10.1

The computer's memory at a particular time is shown in the figure 10.1. The memory frames occupied by the *operating system*, the *fileReader process* and the *average process* are indicated on it by OS, FP and AP respectively.

Selecting from OS, AP and FP, write down the most likely place where each of the following is stored.

(i) content of variable A of the *fileReader process*

(ii) the Process Control Block (PCB) of the *average process*

(*c*) Of the above two python processes, one of them will go through the RUNNING → BLOCKED state transition more than the other. Which process is that? Give the reason for it.

(*d*) Assume that when the *fileReader process* of (b) above is in progress a context switch occurs and a different process is run. When the *fileReader process* is given the chance to run again, the file is read from where it stopped. Which data structure facilitates that feature?

(*e*) A computer uses 32-bit virtual addresses. This computer has a 1 GB (2^{30} bytes) physical memory and a 4 KB page size.

(i) Write down the number of frames in physical memory as a power of 2.

(ii) Assume that in addition to memory frame information, each page table entry for a virtual page in this computer contains some additional information consisting of a total of **four bits**. If the total size of the page table required for each process on this computer assuming that all virtual pages are in use is given as $2^p \times q$ bits, write down the values of p and q.

(iii) If the virtual address 4097 of a particular process is mapped to *Frame* 2 of physical memory, write down in **decimal** form, the physical address corresponding to the virtual address 4097. (Assume that page numbers, frame numbers and addresses begin from 0)

(*f*) The *test.py* file is stored on blocks 218 and 220 respectively in a disk that uses a File Allocation Table (FAT) to manage its storage. The disk uses 4 KB blocks.

(i) Write down an important number in the *directory entry* for the *test.py* file that will help the operating system to find the blocks of the file.

(ii) Give an example size for *test.py* that will result in *internal fragmentation*.

(iii) Assume that block 219 is also to be added for the *test.py* file. Show in a diagram the FAT entries for the *test.py* file after this addition.
(-1 indicates last block)

* * *

5 Paper II mark scheme

Notes
1. Essential keywords sufficient for credit in some answers are underlined.
2. Acceptable alternatives for a given word or set of words are separated by slashes.
3. ←-- *A* indicates that any credit for the item should be given only if A is correct.
4. Answers where *minor* spelling mistakes are acceptable are indicated. A minor spelling mistake is where *at most one character* is either missing, wrong or in excess.
5. Rounding off of 0.5 marks should only be done to the **final total** for Paper II.

1. (a) Draw the expected output of the given HTML code. **[1]**

Cricket
Football
Hockey

* Should be [left aligned / left Justified] [no bullets]

NOTE:

★ Ignore minor spelling mistakes.

★ Ignore case defects. ([illegible])

* Should be left justified and displayed in the left half of the displaying area

(b) Choose suitable number replacements for A to N. **[7]**

0.5 marks for each

A - 7	B - 8	C - 14	D - 1	E - 18	F -15	G - 5
H - 16	I - 4	J - 17	K - 11	L - 20	M -2	N - 12

NOTE: Number should be there. [No wordings allowed]

▼ If the same number is used in more than one occasion, do not give marks for any of them. [for the sections where the same answers are]

(c) Write down the purpose of section P and the purpose of section Q. [2]

1 mark for each:

$ mandatory

P - Get the data entered to the form to variables $name etc.
(without even examples give relevant mark).

Q - Build the SQL query using those variables (+-- *P*)

විමසුම/Query ගොඩනැගීම. (executing SQL) x

2. (a) Write down the second and third steps of the data life-cycle. **[1]**

0.5 marks for each:

2nd: management of data
3rd: removal of obsolete data

(b) (i) Cloud computing model used for storage of data for AI. **[1]**

යටිතල පහසුකම් සේවාවක් ලෙස
Infrastructure as a service / IaaS

NOTE:

★ Ignore minor spelling defects.
★ Ignore case defects.

(ii) Make a suggestion to make the power of quantum computers available for general public. **[2]**

providing the use of quantum computers as a cloud service to the interested users / making it available through the cloud via Infrastructure as a Service (IaaS)

The above total mark is decided as follows:

2 marks if the answer is complete as given above
1 mark if the student has some idea but has not given an answer that is worth full marks

(c) Write down the numbers of the replacements for the blanks. **[2]**

(i) 7 (ii) 3 (iii) 6 (iv) 8 (v) 4

The above total mark is decided as follows:

2 marks if all four or five correct
1 mark if two or three are correct
(if only 1 correct No marks.) [0 marks for duplicated answers] All Zero

NOTE:

▼ If the same number is used in more than one occasion, consider all of them as wrong.

(d) **(i)** Explain what *digital divide* is. **[2]**

Any one from the following:

- economic and social inequality with regard to access to, use of, or impact of information and communication technologies
- the gulf between those who have ready access to computers and the internet, and those who do not
- the gap between demographics and regions that have access to modern information and communications technology (ICT), and those that do not or have restricted access
- unequal access to digital technology, including smartphones, tablets, laptops, and the internet
- the gap between people who have access to modern information and communications technology and those who do not
- the distinction between those who have internet access and are able to make use of new services offered on the World Wide Web, and those who are excluded from these services
- the gap between those with Internet access and those without it

The above total mark is decided as follows:

2 marks if the answer is complete as given above
1 mark if the student has some idea but has not given an answer that is worth full marks

(ii) Write down a step to follow to reduce the environmental impact of our e-waste. [2]

Any one from the following: 3R - 1

- Reduce the unwanted/extravagant use of electronic equipment (e.g., if one already has a working electronic item, it is good for him/her to not to buy a another one)
- Reuse electronic equipment as much as possible (e.g., a broken computer should be repaired if possible)
- Without throwing electronic items that cannot be repaired to garbage dumps, recycle them/their parts to other uses
- If the electronic items have to be disposed, give them to designated e-waste recycling locations
- Without buying new items buy refurbished items
- Extend the lifespan of items by protecting them, using proper maintenance practices, keeping them clean, regularly updating software, protecting them from virus attacks, using protective cases, using screen protectors, using surge protectors etc
- Donate or sell unwanted electronics
- Go for digital minimalism (e.g.,decluttering and organizing files, reducing unnecessary downloads and backups, and deleting unused applications)

The above total mark is decided as follows:

2 marks if the answer is complete as given above
1 mark if the student has some idea but has not given an answer that is worth full marks

3. (a) Write down the replacements for A to G in the flowchart. **[3]**

Marks allocated as follows:

AB 0.5 marks: A: no B: yes

C 0.5 marks: sum = sum + count
sum += count ——— (←-- *AB*)

ALTERNATIVE 1:

E 1 mark: count ≤ 18? count count ≤ 20?
count < 19? — count count < 210 ? — (←-- *C*)

? complusory

DF 0.5 marks: D: yes F: no ——— (←-- *E*)

ALTERNATIVE 2:

E count >= 18 ?
E count >= 21 ?

E 1 mark: count > 18? — count > 20? count ≥ 19? — (←-- *C*)

DF 0.5 marks: D: no F: yes ——— (←-- *E*)

G 0.5 marks: print sum / output sum / display sum / show sum — (←-- *DF*)

NOTE:

▼ For the E condition, the question mark symbol (?) is essential.

★ For C and E, correct textual descriptions are also acceptable.

★ Ignore case.

(b) (i) What is the output of the given Python code? **[2]**

```
[4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30]
```

Marks allocated as follows:

A: **1.5 marks** for correct list content

B: **0.5 marks** for [] and commas (←-- *A*)

NOTE:

★ Ignore space defects.

(ii) What would be the output if i%2==0: is replaced with i%2!=0: ? **[2]**

```
[5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29]
```

Marks allocated as follows:

A: **1.5 marks** for correct list content
B: **0.5 marks** for [] and commas (←-- *A*)

NOTE:

★ Ignore space defects.

(c) Write down the replacements for the labels in the python code to find the largest in a set of numbers. **[3]**

0.5 marks for each:

A: Any negative value / 0 / 1 / 2 / 3 / 4 / myList[0]
B: myList (←-- exact spelling,case)
C: largest: (←-- A, B, exact spelling,exact case, colon)
D: i (←-- *C*, exact case)
E: largest in largest :; must be there and ; (←-- D, exact spelling,exact case)
F: list1 (←-- exact spelling, exact case)

4. (a) Write down the numbers of the replacements for DFD labels. [4]

0.5 marks for each:

P: 5
Q: 4 (←-- *P*)
R: 2 (←-- *Q*)
S: 3
T: 6
U: 7 (←-- *P*)
V: 8 (←-- *U*)
W: 1

NOTE:

(for all the repeating answers 0)

▼ If the same number is used in more than one label, consider all of them as wrong.

(b) Write down one functional requirement w.r.t. reservations. [1]

Any one from the following:

Even the answer is split & Answer is read & correct give marks

- Authentication of student when he/she logs into the system
- Computers to be available for reservation for weekend 30 minute time slots between 8am and 5pm / computers to be available for reservations for weekends. / computers to be available for reseravation for weekends 30 points time slots.
- A student to be given a maximum of two 30 minute time slots
- One computer to be reserved by only one student for a particular time slot

(c) Write down one thing to do when checking technical feasibility of the project. [1]

Any one from the following:

- whether it is technically possible to do the project
- whether it is possible to develop the product with the available technology in the school
- whether it is possible to add more technical resources if needed
- whether the chosen technology is the right choice to help the team complete the system within the budget and time allocated for the project or whether there are other better choices
- whether the school requires specific technology, or is the school open to developing the product, irrespective of the technology
- whether open source software could be used
- whether the technical resources (software/hardware) are available
- whether the technical resources are adequate
- whether the technical team is capable to make a working system
- whether the system should be compatible with other existing systems in the school

(d) Why a proper requirement analysis is critical for the timely completion of this project which uses the waterfall model? [1]

In the waterfall model, each phase must be completed before the next phase can begin. Thus if the requirement analysis is not properly done during its phase then it is likely that a system that will be rejected by the customers will be developed. It will be very costly and time consuming to correct that mistake.

(e) What is *integration testing* in this example? [1]

After the unit testing of the three modules are done, then the modules need to be integrated into a single system. Testing this single system as a combined unit is done during integration testing.

(f) Why didn't the teacher suggest a *parallel deployment*? [1]

Any one from the following:

- more resources are needed for it
- it is more time consuming to maintain two systems
- it is more costly to maintain two systems
- it takes effort to keep the two systems consistent
- it can cause confusion and frustration
- as it is for school's own use, one need not take the effort of parallel deployment as any mistakes can be remedied comparatively easily
- this is not a high risk system that warrants a parallel deployment

(If advantages of other deployments are writen → X marks)

(g) Why should a COTS for this system be rejected? [1]

Any one from the following:

- it may not fit the school requirements
- customization restrictions
- vendor dependence
- integration challenges
- may have a higher cost
- the need for substantial training
- students missing a learning opportunity
- may not have Sinhala, Tamil language support
- may have unnecessary features
- may have licensing costs
- can be more expensive over time
- can be impossible or inflexible to change if one needs it
- school may not have control
- may not be supported after some time
- upgrades can cost extra

- it may not have the exact features the school needs
- might not fit school's work processes

5. (a) (i) Draw the complete truth table for the required circuit. [3]

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

The above total mark is decided as follows:

3 marks for all 8 rows correct
2.5 marks for maximum 6,7 rows correct
2 mark for maximum 3,4,5 rows correct
1 mark for maximum 1,2 rows correct

NOTE:

★ Having *Output* as the Z column title is acceptable.

▼ If the Z column is not labelled, or the label is different from Z / *Output*, **reduce 1 mark** from the earned total.

(ii) Complete the Karnaugh map according to the given format. **[2]**

01
~~0.25~~ marks for each correct ~~cell~~: Row

ALTERNATIVE 1:

C \ AB	00	01	11	10
0	0	0	1	0
1	1	1	0	0

ALTERNATIVE 2:

C \ AB	00	01	11	10
0	0	0	1	0
1	1	1	0	0

no need groupings

NOTE:

▼ For both alternatives, indicating all 1's and 0's are compulsory for credit.

the 0-groupings although drawn here is required for part (iii)

the is requar

(iii) Using the K map, derive a simplified POS expression for X. [3]

Simplified POS expression for ALTERNATIVE 1:

groupings compulsory

$$Z = (A' + C')(A + C)(B + C)$$

Simplified POS expression for ALTERNATIVE 2:

$$Z = (A + C)(A' + C')(A' + B)$$

Marks allocated as follows:

A: **1.5 marks** for marking the three loops on the correct Karnaugh map (**0.5 marks** for each)

B: **1.5 marks** for correct, simplified final POS expression for the used alternative

NOTE:

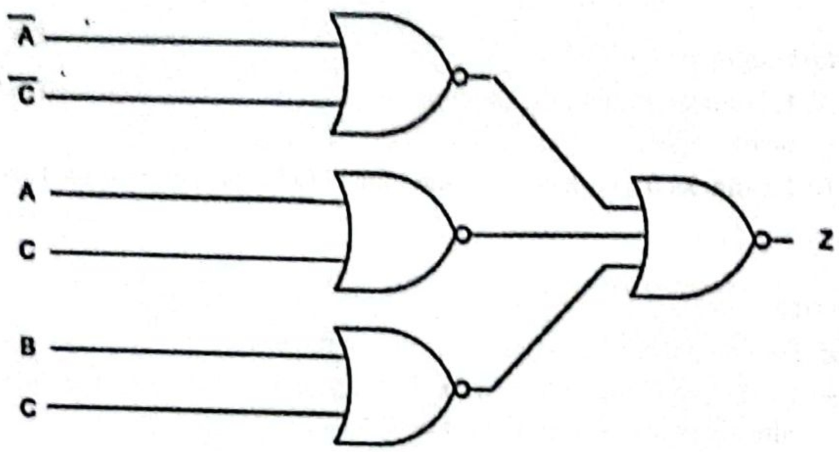
★ For component **B**, the term **Z** is not compulsory.

★ 1 cells not being indicated on the Karnaugh map is permissible as the student has already being penalized for it in Part (ii).

(iv) Draw a logic circuit for the above simplified expression by only using NOR gates. [2]

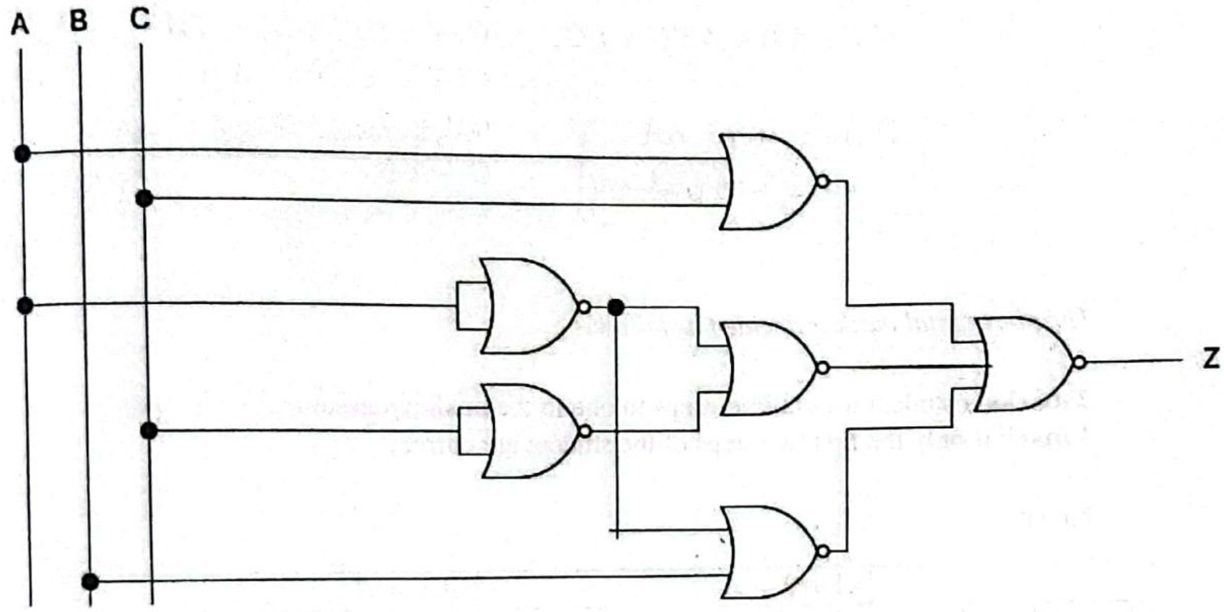
FOR ALTERNATIVE 1:

$$Z = (A' + C')(A + C)(B + C)$$



FOR ALTERNATIVE 2:

$$Z = (A + C)(A' + C')(A' + B)$$



Marks allocated as follows for the drawn alternative:

A: **1 mark** for the first set of NOR gates (←-- Correctly simplified expression for Z)

B: **1 mark** for the final NOR gate (←-- *A*)

NOTE:

▼ If the wire connections are not clearly indicated on on a correct circuit, then give only a maximum of **1 mark**. The student can either indicate the wire connections using the dark dots (as shown in the diagram or use half-circles) to indicate non-connecting wires.

★ The **Z** term is not compulsory.

* Complimented inputs canbe input using NOR gates in altertive parts. Similerly complemented inputs can be obtained directly in alternative 2.

(b) Using Boolean algebra, show that $A'C + A'B + AB'C + BC$ is equivalent to $C + A'B$. **[2]**

$$
\begin{aligned}
A'C + A'B + AB'C + BC &= C(A' + AB' + B) + A'B \\
&= C(A' + A + B) + A'B \\
&= C(1 + B) + A'B \\
&= C + A'B
\end{aligned}
$$

[Rules are not compulsory]

The above total mark is decided as follows:

2 marks if student uses correct steps to obtain the final expression
1 mark if **only** the first two steps of the student are correct

NOTE:

	$A + 0 = A$	$A.1 = A$
Inverse/Complement	$A + A' = 1$	$A.A' = 0$
	$A + A = A$	$A.A = A$
Identity	$A + 1 = 1$	$A.0 = 0$
Involution	$(A')' = A$	
Commutative	$A + B = B + A$	$AB = BA$
Associative	$A + (B + C) = (A + B) + C$	$A(BC) = (AB)C$
Distributive	$A(B + C) = AB + AC$	$A + BC = (A + B)(A + C)$ Derivative: $A + A'B = A + B$
DeMorgan	$(A + B)' = A'B'$	$(AB)' = A' + B'$
Absorption	$A + AB = A$	$A(A + B) = A$

Table 1: Postulates and theorems of Boolean Algebra

(c) (i) Output at Q when S=1 and R=0? [1]

1

(ii) Q when S is now made 0? [1]

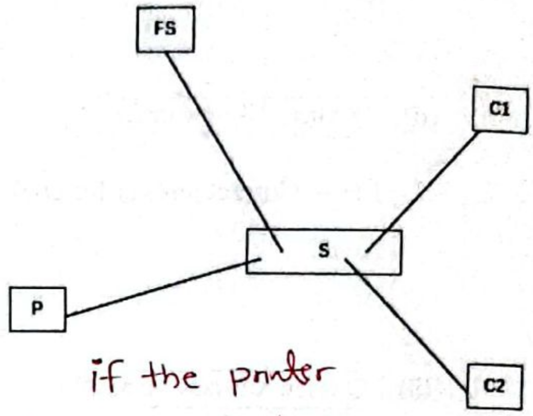
1 (←-- Correct answer for c(i))

(iii) Q when R is now made 1? [1]

0

6. **(a)** Sketch how a FS, S, P, C1 and C2 should be connected in a star topology.

[1]



if the printer is connected to a computer give marks.

The printer could be connected to one of the computers as well.

(b) The use of the port number in an Internet connection?

it identifies the service point

[1]

It identifies the process that is relevant to the connection.

(It identifies the process.)

ක්‍රියාවලිය හඳුනා ගනී

Identify correct application / x Software

(c) **(i)** Write an example IP address that can be assigned to a host attached to this subnet.

[1]

An example IP address is 192.168.56.138

Any answer that lies between 192.168.56.129 and 192.168.56.190 (both inclusive) is also acceptable.

(ii) Write the first and the last usable host addresses in this network. [1]

0.5 marks for each:

first address: 192.168.56.129
last address: 192.168.56.190

(iii) How many host addresses are available for use in this subnet? [1]

62

if 61 is written must what happened to other 1

(d) (i) Write the subnet mask of the given IP address block in dotted decimal notation. [1]

255.255.255.192

(ii) Write the number of host bits needed to create the required number of subnets. [1]

2

(iii) Fill the table. [4]

ALTERNATIVE 1:

Give **1 mark** for each correct row.

Subnet	Network Address	First Usable IP address	Last usable IP address	Broadcast address
A	192.168.56.0	192.168.56.1	192.168.56.14	192.168.56.15
B	192.168.56.16	192.168.56.17	192.168.56.30	192.168.56.31
C	192.168.56.32	192.168.56.33	192.168.56.46	192.168.56.47
D	192.168.56.48	192.168.56.49	192.168.56.62	192.168.56.63

ALTERNATIVE 2:

Give **1 mark** for each correct row.

Subnet	Network Address	First Usable IP address	Last usable IP address	Broadcast address
A	192.168.56.32	192.168.56.33	192.168.56.38	192.168.56.39
B	192.168.56.40	192.168.56.41	192.168.56.46	192.168.56.47
C	192.168.56.48	192.168.56.49	192.168.56.54	192.168.56.55
D	192.168.56.56	192.168.56.57	192.168.56.62	192.168.56.63

(e) (i) Write two functions of a proxy server in a computer network. [2]

Any **two** from the following with **1 mark** for each:

- acting as an intermediary between the user's computer and the Internet / offering access to uncensored Internet / allowing client computers to make indirect network connections to other network services / helping users browse the web anonymously / providing a high level of privacy / hiding actual IP addresses of users

- storing recently requested web objects/pages for future requests
- reducing the time required to access web pages because of caching
- hiding a network from outside and thus securing that network / preventing attackers from entering a private network
- forwarding web requests
- content filtering
- acting as a firewall
- saving network bandwidth / improving network performance / network connection sharing
- helping control Internet usage of users

(ii) Write two properties of MAC addresses assigned to devices connected to a network. [2]

Any **two** from the following with **1 mark** for each:

- They are 48 bits in length. / They are divided in to six blocks separated by colons. / It is a six byte hexadecimal address. / It is a 48-bit address that contains six groups of two hexadecimal digits separated by colons.
- They are physical addresses. / MAC address is hardware oriented. / They are hard-coded into the device. / They are attached to the network interface (host).
- They are assigned by the manufacturer.
- They are permanent. / They cannot be changed.
- They are unique addresses assigned to interfaces of a host./ MAC address sharing is not allowed.
- When data is sent, MAC addresses enable the unique identification of the device interface. / They uniquely identify the devices on a network. / MAC addresses support the correct delivery of the data to the receiver's interface. / A switch needs the MAC address to forward data.
- MAC address operates in the data link layer.
- MAC addresses cannot be found easily by a third party.

7. (a) (i) Identify the parts marked as A,B,C and D and briefly explain their functionalities. **[4]**

0.5 marks for naming and **0.5 marks** for describing each function;
($0.5 \times 2 \times 4 = 8$ marks)
4

A – USB Port -
The function could be any one of the following:

- Could be used to connect a computer to the Board
- Could be used to upload firmware into the micro-controller
- Could be used for data communication between the computer and the Board

compulsory

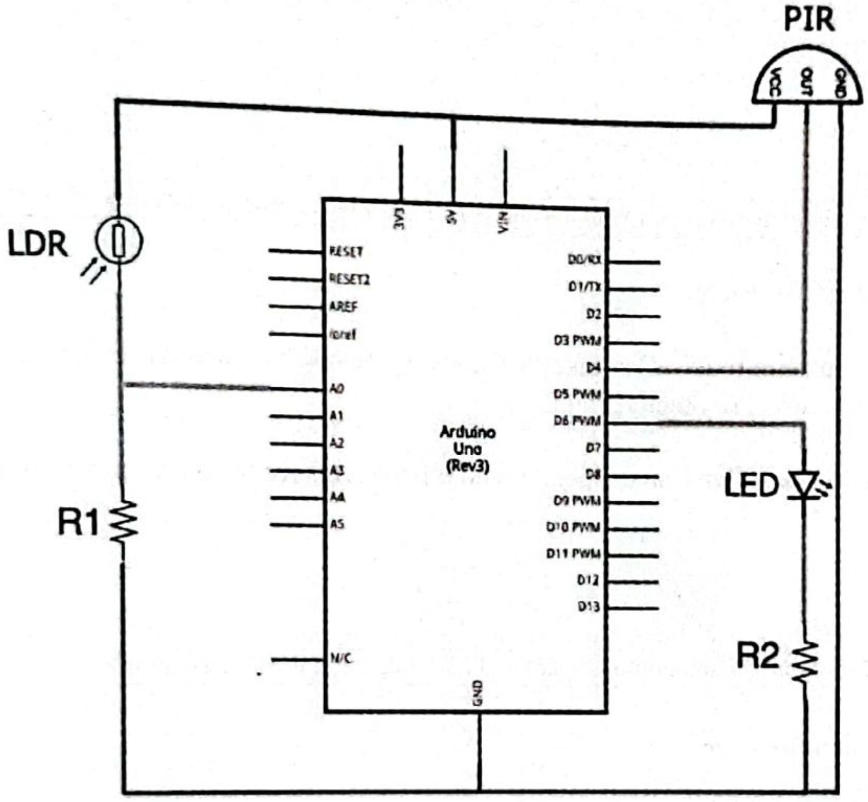
B – Analog input pins – Feed analog inputs to the micro-controller

C – Micro-controller – Any answer that clearly explains the task of processing inputs to Board and producing digital output based on the computations carried out

compulsory

D – Digital input/output pins – Feed digital inputs as well as deliver digital outputs.

(ii) Draw a schematic diagram for the required IOT setup. [3]



Allocate marks as follows:

A: **1 mark** for the use of resistors R1 and R2 and correct ground/5V connections
B: **1 mark** for correct LDR connection to analog pins and LED to a digital pin
C: **1 mark** for correct PIR sensor output to another digital pin

NOTE:

★▼ Complete pin details are not needed, yet connecting pins must be identified.

★ Connecting the output of the PIR to an analog pin is also acceptable.

★ LDR sensor can also be indicated as the "sensor for ambient light detection"

★ Switching the positions of LED and R2 is accepted.

(b) (i) Briefly explain (a) autonomous and (b) cooperative characteristics of a software agent. [2]

1 mark for each:

(a) Autonomous – Can take decisions by themselves without being controlled by the others (users or other agents)

(b) Cooperative Can cooperate with other agents (or users) when performing tasks

(ii) Identify a self-autonomous agent and a user agent in the given example. [2]

1 mark for each:

Self autonomous agent: multi-agent robots

User agent: either Delivery Handler Agent OR Dispatch Handler Agent

(iii) Write down the most likely observation when the multi-agent robots satisfy only the autonomous characteristic but fail to cooperate. [1]

There will be competitive behaviour among multi agent robots to complete their tasks which they try to complete individually. E.g., Loading area will be congested by the competing robots to pick the next package. (or each multi agent robot will see all other multi agent robots as obstacles to their work)

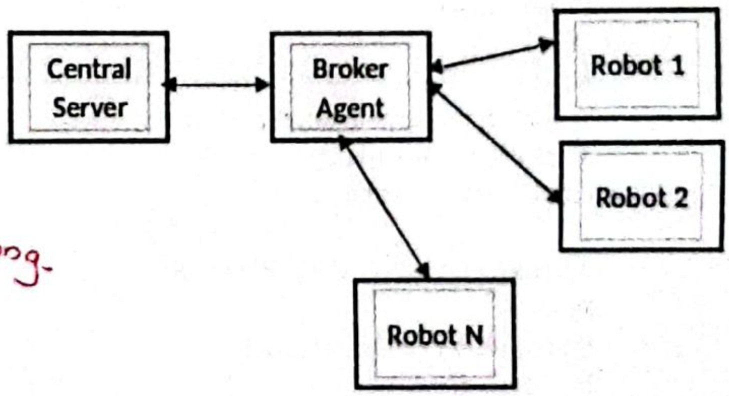
(iv) It the system is redesigned by replacing the multi-agent behaviour with centralized control and a broker agent for communication, identify one main change with respect to (a) control of the robot mobility and (b) decision making process relevant to moving packages from loading to dispatch areas. [2]

1 mark for each:

(a) Each robot will only move based on the mobility instructions received from the central server.

(b) The central server receives data, processes them at the server and instructs the robots to pick up the packages, move through the package moving area with respect to their dispatch assignments. Communication is facilitated through the broker agent.

(v) Draw a *box and arrow diagram* for the new solution with centralized control. [1]



Single arrow head is wrong.

NOTE

Connecting components with lines (Without arrow heads) are acceptable. But if the arrows are used then each connection must contain 2 arrow heads.

8. (a) Write down the output of the Python code of Figure 8.1. **[2]**

If the 'question' is wrong written give 2 marks

L*br*ry

NOTE:

▼ Exact answer with proper case required.

ALTERNATIVE: If the student's answer is "error" due to quotes in code being misinterpreted, just give 1 mark.

(b) Write down the suitable replacements for P-U in the bubbleSort function. **[3]**

0.5 marks for each:

P: len(nList)-1
Q: 0
R: -1 (←-- *P, Q*)
S: for i in range(pNumber) (←-- *R*)
T: nList[i] = nList[i+1]
U: nList[i+1] = temp

ALTERNATIVE FOR P, Q, R AND S:
P: 0
Q: len(nList) OR len(nList)-1
R: 1 (←-- *P, Q*)
S:

for i in range(0, len(nList)-1) OR

for i in range(len(nList)-1) OR

for i in range(0, len(nList)-1-pNumber) (←-- *R*)

NOTE:

▼ Exact spelling and case required.

(c) (i) Write down the suitable replacements for the labels A-J of the Python code in Figure 8.3. [8]

A: 0.5 marks open
B: 0.5 marks not OR "" == OR '' ==(with no space between quotes)
C: 0.5 marks break
D: 1 mark empDetails[1]
E: 1 mark topay//notes[i]
F: 1 mark required[i] (←-- *E*)
G: 1 mark topay%notes[i] (←-- *E*) topay - required [i] * notes [i]
H: 1 mark i = i + 1
I: 1 mark required[i] (←-- *E*)
J: 0.5 marks file.close() (←-- ~~*P, Q, R*~~ A)

NOTE:

▼ Exact spelling and case required.

(ii) The problem in code with respect to net pay inputs? What will you do to fix that problem? [2]

1 mark for *problem* and **1 mark** for *solution*:

Problem: Can only handle net pay inputs that can be made with the notes in the notes array. E.g., It cannot handle a net pay input like 40001 or 40010.

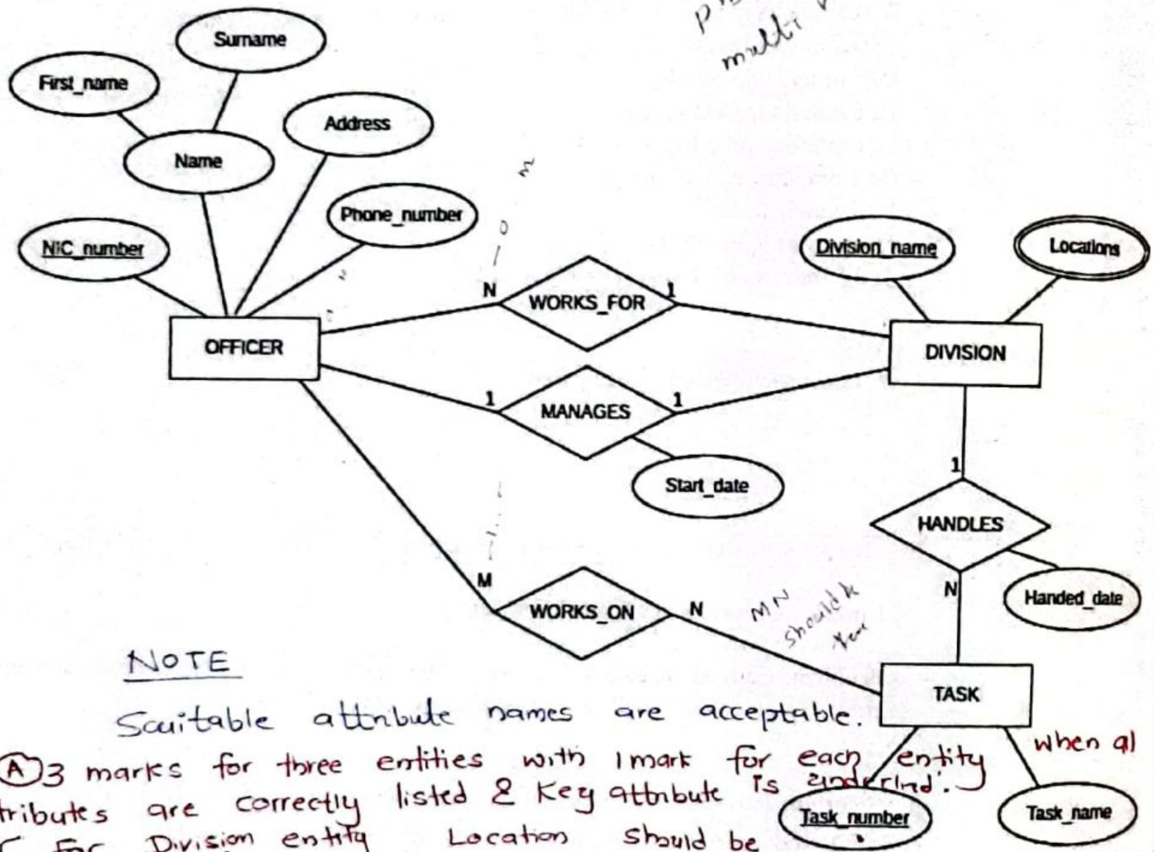
Solution: Two solutions. Either one is acceptable.

- Do not process if a net pay input is not divisible by either 50 or 20.

- Increase the size of the arrays as follows:
  notes = [5000,1000,500,100,50,20,10,5,2,1]
  totals = [0,0,0,0,0,0,0,0,0,0]
  required = [0,0,0,0,0,0,0,0,0,0]

9. (a) Draw the ER diagram for the office database. [5] 8

phone num can be multi valu + assumptions.



NOTE

Suitable attribute names are acceptable.

(A) 3 marks for three entities with 1 mark for each entity when all its attributes are correctly listed & Key attribute is underlined.

[For Division entity Location should be multivalued)

(B) 4 marks for the four relationships with 1 mark for each.

Marks allocated as follows: (C) 1 mark for the two relationship attributes. (0.5 for each)

A: **1 mark** for the three entities (with the key attribute underlined) connected through (correct or incorrect) relationships

B: **1 mark** for indicating the *Locations* multi-valued attribute (←-- *A*) connected properly for DIVISION.

C: **1 mark** for the four relationships with correct/incorrect cardinality (←-- *A*)

D: **1 mark** for indicating correct cardinality (←-- *C*)

E: **1 mark** for all attributes correctly being listed and connected properly (←-- *D*) A, C

(b) Write two advantages of converting a database into a normal form. **[1]**

Any **two** from the following with **0.5 marks** for each:

- minimizes the physical storage space required / reduces redundant data / reduces data duplication
- increases data integrity / provides data consistency / prevents data anomalies
- supports efficient query response
- provides for a more flexible database design
- increases database security
- provides better and quicker execution
- making updates to data is easier because it need not be done in multiple places / easy to maintain
- references to a record can be changed without removing the record
- reduces the risk of data entry errors

(c) (i) In which normal form does the Show table exist? Justify your answer. **[2]** 1

~~**1**~~ 0.5 **mark** for the normal form and ~~**1**~~ 0.5 **mark** for justification:

Second Normal Form / 2nd / 2NF

ts

Justification: (←-- Correct normal form [2NF])

Any **one** from the following:

k for each.

- It is in 1NF and every field that is not part of the primary key is functionally dependent on the whole of the primary key.

. (0.5 for each)

- It is in 1NF and no partial dependencies. Therefore, 2NF.
- It is in 1NF and is not in 3NF due to the transitive dependency. Therefore, 2NF.

(ii) Convert the **Show** table to its next normal form. 1 ~~[2]~~

0.5
~~1~~ **mark** for each:

A: Show(<u>Theatre</u>, <u>Day</u>, <u>Time</u>, <u>Screen</u>, Movie)

B: Movie_Year(<u>Movie</u>, Year) ~~(← A)~~

NOTE: 0.5

▼ Reduce ~~1~~ **mark** from total if primary keys not underlined / spelling defects / case defects.

★ Student can provide the relation names.

(d) (i) Write the SQL statement to create the **Employee** table with a suitable primary key. ~~[2]~~ 1

```
CREATE TABLE Employee (
  Emp_ID VARCHAR(4) PRIMARY KEY,
  Emp_Name VARCHAR(50),
  DoB DATE,
  Department VARCHAR(50),
  Designation VARCHAR(50),
  DoJ DATE,
  Salary DECIMAL(10,2)
);
```

ignore ;

ALTERNATIVE:

```
CREATE TABLE Employee (
  Emp_ID VARCHAR(4),
  Emp_Name VARCHAR(50),
  DoB DATE,
  Department VARCHAR(50),
  Designation VARCHAR(50),
  DoJ DATE,
  Salary DECIMAL(10,2),
  PRIMARY KEY (Emp_ID)
);
```

Acceptable alternative data types:
Emp_ID CHAR(4)
Salary INT
Salary FLOAT(10,2)

Marks allocated as follows:

A: 0.5 ~~1~~ **mark** for correct CREATE TABLE Employee(Exact field names); (←-- The semi-colon, exact spelling and case of field names)

B: 0.5 ~~1~~ **mark** for choosing Emp_ID as the primary key AND the correct data type usage ~~(←-- A)~~ (←-- A).

NOTE:

★ Student can choose the field sizes.

(ii) Write the SQL statement to insert the given employee record. [1]

```
INSERT INTO Employee (Emp_ID, Emp_Name, DoB, Department, Designation,
DoJ, Salary)
VALUES ('E119', 'John', '15-06-1971', 'IT', 'Professor', '15-07-2001', 107000);
```

x Do not ignore ';'

ALTERNATIVE: Field names can be omitted but the values for all columns must be there as shown below:

```
INSERT INTO Employee  VALUES ('E119', 'John', '15-06-1971', 'IT', 'Professor',
'15-07-2001', 107000);
```

INSERT INTO employee (Emp-ID, Emp-Name, DOB, Department, Designation, DOJ, Salary) VALUES ('0002', 'John Doe', '1971-06-15', 'IT', 'Developer', '2001-07-15', 75000);

NOTE:

▼ The semicolon, exact spelling and case of table name and the field names are required.

★ Ignore minor spelling mistakes of the inserted data values.

(iii) Write the output of the given SQL query. [1]

Emp_ID	Emp_Name
E110	Saman
E114	Jennifer
E119	John

NOTE:

▼ Exact spelling and case of field names and values are required.

★ Writing only the entries for E110 and E114 (without the one for E119) is also acceptable .

(iv) Write the SQL query to find the names of all employees who work in the "Civil" department. **[1]**

```
SELECT Emp_Name
FROM Employee
WHERE Department = 'Civil';
```

Alternative answer:

```
SELECT Emp-Name
FROM Employee
WHERE Department LIKE
'Civil';
```

NOTE:

▼ The semicolon, exact spelling and case of field names and the text (Civil) are required.

* Double quotes (" ") are acceptable.

10. (a) (i) What is the repeating cycle that a processor in a computer is involved in? [1]

fetching instructions - decoding and executing them / Fetch-Decocde-Execute / Fetch - execute

(ii) Which program's instructions get executed during a *context switch*? [1]

the operating system's

(iii) How many flip-flops are needed to create an *n* bit register? [1]

n

(b) (i) Where is the content of variable A in the *fileReader process* stored? [1]

FP fp area

(ii) Where is the PCB of *average process* stored? [1]

OS

(c) Which process goes through the RUNNING → BLOCKED transition more than the other? Give the reason. [2]

1 mark for each:

A fileReader

B As there is file reading in it (← 2A).

As it is Involved in I/O

(d) Which data structure facilitates reading a file from its stopped position? [1]

PCB of the fileReader process

if only (PCB) is not acceptable.

(e) (i) Write down the number of frames in physical memory as a power of 2. [1]

2^{18}

$\frac{2^{30}}{\text{Pg size}}$ ✓ → $\frac{2^{30}}{2^{12}} = 2^{18}$

(ii) Write down the values of p and q. [2]

1 mark for each:

p = 20

q = 22

(iii) Write down in decimal the physical address corresponding to the virtual address 4097. [1]

8193

(f) (i) Write down an important number in the directory entry for *test.py* that will help the OS to find its blocks. [1]

218

(ii) Give an example size for test.py that will result in *internal fragmentation.* [1]

Any size between 4 - 8 KB

4, 8 ×

5, 6, 7

Any size with

219

(iii) Show the FAT entries for test.py after block 19 is added. [1]

order need to be there.

218	220
219	-1
220	219

NOTE:

▼ Blocks from 218 should be drawn in sequence.